Kinematic Distributions in the $Z \to \ell\ell\gamma$ Photon Shower Shape Study: MC Normalisation, Data–MC Agreement, and Conversion Rate Discrepancies

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Abstract

This note documents the investigation of photon kinematic distributions (p_{T} and $|\eta|$) in $Z \to \ell\ell\gamma$ events selected from 108 fb^{-1} of ATLAS data collected at $\sqrt{s} = 13.6$ TeV in 2024, compared to Sherpa MC23e Monte Carlo simulation. A significant normalisation discrepancy was traced to the use of DAOD-level sum-of-weights in the NTupleMaker framework, which misses events removed by derivation skimming. Replacing these with EVNT-level values from AMI resolved the over-estimation. The remaining ∼20% normalisation deficit is attributed to missing Z+jets background and incomplete MC sample coverage. Shape comparisons between data and MC reveal good agreement in p_{T}, while $|\eta|$ distributions show conversion-status–dependent discrepancies linked to a known TRT barrel tracker issue in Run 3.

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1 Introduction

The photon shower shape reweighting analysis requires precise data–MC comparisons of photon properties in $Z \to \ell\ell\gamma$ events. As a prerequisite, the kinematic distributions — photon transverse momentum p_{T} and pseudorapidity $|\eta|$ — must demonstrate adequate agreement between data and simulation, both in shape and in overall normalisation.

This study was motivated by the observation that the MC prediction significantly exceeded the data yield. The investigation revealed a normalisation bug in the MC event weighting, related to the handling of the sum-of-weights in the presence of DAOD-level skimming. After correcting this, residual differences were found in the $|\eta|$ distributions that depend on the photon conversion status, pointing to a known issue with the TRT barrel tracker in Run 3.

2 Samples and Event Selection

2.1 Data

The analysis uses 108 fb^{-1} of proton–proton collision data collected by the ATLAS detector in 2024 at $\sqrt{s} = 13.6$ TeV. Events are selected from two DAOD derivations:

- **EGAM3**: $Z \to ee\gamma$ candidates,
- **EGAM4**: $Z \to \mu\mu\gamma$ candidates.

The two channels are analysed separately and also combined into an inclusive $Z \to \ell\ell\gamma$ sample.

2.2 Monte Carlo

Simulated $Z \to \ell\ell\gamma$ events are generated with Sherpa 2.2.14 using the MC23e campaign. Two samples are used:

Table 1: MC samples and normalisation parameters.

DSID	Process	σ [pb]	AMI EVNT sumW	Total events
700770	Sh_2214_eegamma	102.60	5.269×10^{14}	185 402 000
700771	Sh_2214_mumugamma	102.59	5.421×10^{14}	189 985 000

Cross-sections are obtained from the PMG cross-section database (`PMGxsecDB_mc23.txt`). The generator filter efficiency is 1.0 for both samples.

2.3 Event Selection

The baseline selection follows Francisco's configuration for the cell-energy reweighting study:

Table 2: Event selection criteria.

Variable	Requirement
Photon p_{T}	> 10 GeV
Photon $\lvert\eta\rvert$	< 2.37, excluding crack $[1.37, 1.52)$
$m_{\ell\ell}$	$[40, 83]$ GeV
$m_{\ell\ell\gamma}$	$[80, 100]$ GeV
$\Delta R(\gamma, \ell)$	> 0.4 (both leptons)

Several scenarios are studied, varying the photon conversion status and isolation requirement:

Table 3: Selection scenarios.

Scenario	Conversion	Photon ID	Isolation
Baseline (unconverted)	Unconverted only	No ID	Loose
Converted	Converted only	No ID	Loose
All conversions	Both	No ID	Loose
Tight isolation	Unconverted only	No ID	Tight

3 MC Event Weight and the Normalisation Problem

3.1 MC Weight Formula

Each MC event is weighted by:

$$w = w_{\mathrm{mc}} \times w_{\mu} \times \frac{\sigma \times \mathcal{L}}{\sum w_{\mathrm{EVNT}}} \times \mathrm{SF}_{\ell_1} \times \mathrm{SF}_{\ell_2} \times \mathrm{SF}_{\gamma,\mathrm{iso}} \tag{1}$$

where w_{mc} is the generator weight, w_{μ} the pileup reweighting factor, σ the cross-section, $\mathcal{L} = 108\ \mathrm{fb}^{-1}$ the integrated luminosity, $\sum w_{\mathrm{EVNT}}$ the EVNT-level sum of generator weights, and the SF terms are lepton and photon isolation scale factors.

An alternative weight without scale factors is also computed for comparison:

$$w_{\mathrm{no\,SF}} = w_{\mathrm{mc}} \times w_{\mu} \times \frac{\sigma \times \mathcal{L}}{\sum w_{\mathrm{EVNT}}} \tag{2}$$

3.2 The Bug: DAOD-level Sum-of-Weights

The NTupleMaker framework, developed to process input AOD files (which are unskimmed), computes a sum of generator weights and stores it in a histogram (`h_sumW`) in the output ROOT file. The sum-of-weights is retrieved from bin 2 of this histogram.

However, the production ntuples used in this analysis were derived from **DAOD** (Derived Analysis Object Data) files, which are **skimmed**: events that fail the derivation-level selection are removed before reaching the NTupleMaker. Consequently, the recorded sum-of-weights only accounts for events that survived the DAOD skimming — approximately **36%** of the full EVNT-level sample.

Using this DAOD-level $\sum w$ in the denominator of Equation 1 inflates the per-event weight by a factor of:

$$\frac{\sum w_{\mathrm{EVNT}}}{\sum w_{\mathrm{DAOD}}} \approx 2.77 \tag{3}$$

This resulted in the MC prediction overshooting the data by a factor of $\sim$2.2 (MC/Data $\approx$ 2.19), which was immediately visible in the event-count kinematic distributions.

3.3 The Fix: AMI EVNT-level Sum-of-Weights

The fix replaces the dynamically computed DAOD-level sum-of-weights with the correct EVNT-level values obtained from AMI (the ATLAS Metadata Interface). The corrected values are hardcoded in the analysis configuration:

- DSID 700770: $\sum w = 5.269047 \times 10^{14}$
- DSID 700771: $\sum w = 5.421893 \times 10^{14}$

After this correction, the MC/Data ratio drops to $\sim$0.79, i.e. the MC now *under*predicts the data by about 20%.

3.4 Residual Normalisation Deficit

The remaining $\sim$20% deficit has several contributing causes:

1. **Missing Z+jets background**: The MC samples contain only the signal process $Z \to \ell\ell\gamma$. The data inevitably includes Z+jets events where a jet is misidentified as a photon. This background is not subtracted and contributes additional events in data.

2. **Incomplete MC sample coverage**: The available MC samples cover 97.59% of the total generated events; some sub-jobs may have failed or not been included.

3. **Filter efficiencies**: Generator-level filter efficiencies and potential differences in the DAOD derivation acceptance are not fully accounted for.

Crucially, this residual normalisation offset does **not** affect the shower shape reweighting, which uses *area-normalised* distributions. The key requirement is that the *shape* of the kinematic distributions agrees between data and MC.

4 Results: Loose Isolation

Each figure shows the photon p_{T} and $|\eta|$ distributions for data (black markers), MC with scale factors (red histogram), and MC without scale factors (blue dashed histogram). The left column shows absolute event counts and the right column shows area-normalised distributions. A ratio panel (MC/Data) is included below each comparison.

4.1 Unconverted Photons

loose/llgamma/baseline/plots/kinematics_pt.pdf

(a) p_{T} (events)

loose/llgamma/baseline/plots/kinematics_pt_norm.pd

(b) p_{T} (normalised)

loose/llgamma/baseline/plots/kinematics_eta.pdf

(c) $|\eta|$ (events)

loose/llgamma/baseline/plots/kinematics_eta_norm.p

(d) $|\eta|$ (normalised)

Figure 1: Unconverted photons, loose isolation. The p_{T} shapes agree well between data and MC. The $|\eta|$ distribution shows a deficit in MC in the barrel region, particularly at low $|\eta|$.

4.2 Converted Photons

loose/llgamma/converted/plots/kinematics_pt.pdf

(a) p_{T} (events)

loose/llgamma/converted/plots/kinematics_pt_norm.p

(b) p_{T} (normalised)

loose/llgamma/converted/plots/kinematics_eta.pdf

(c) $|\eta|$ (events)

loose/llgamma/converted/plots/kinematics_eta_norm.

(d) $|\eta|$ (normalised)

Figure 2: Converted photons, loose isolation. Notably better data–MC agreement in $|\eta|$ compared to the unconverted case, as discussed in Section 6.2.

4.3 All Conversions

loose/llgamma/all_conv/plots/kinematics_pt.pdf

(a) p_{T} (events)

loose/llgamma/all_conv/plots/kinematics_pt_norm.pd

(b) p_{T} (normalised)

loose/llgamma/all_conv/plots/kinematics_eta.pdf

(c) $|\eta|$ (events)

loose/llgamma/all_conv/plots/kinematics_eta_norm.p

(d) $|\eta|$ (normalised)

Figure 3: All photons (unconverted + converted), loose isolation. The inclusive $|\eta|$ shape shows intermediate agreement, mixing the two sub-populations.

5 Results: Tight Isolation

Applying tight isolation reduces the contribution of Z+jets background, where jets faking photons tend to be less isolated. As suggested by the supervisors, this was investigated to see if tighter selection improves the data–MC agreement.

5.1 Unconverted Photons

iso_tight/llgamma/baseline/plots/kinematics_pt.pdf

(a) p_{T} (events)

iso_tight/llgamma/baseline/plots/kinematics_pt_nor

(b) p_{T} (normalised)

iso_tight/llgamma/baseline/plots/kinematics_eta.pdf

(c) $|\eta|$ (events)

iso_tight/llgamma/baseline/plots/kinematics_eta_no

(d) $|\eta|$ (normalised)

Figure 4: Unconverted photons, tight isolation. The normalisation gap narrows slightly compared to the loose isolation case (Figure 1), consistent with the rejection of Z+jets background in data.

5.2 Converted Photons

iso_tight/llgamma/converted/plots/kinematics_pt_tight

(a) p_{T} (events)

iso_tight/llgamma/converted/plots/kinematics_pt_no

(b) p_{T} (normalised)

iso_tight/llgamma/converted/plots/kinematics_eta_tight

(c) $|\eta|$ (events)

iso_tight/llgamma/converted/plots/kinematics_eta_n

(d) $|\eta|$ (normalised)

Figure 5: Converted photons, tight isolation. The agreement remains comparably good as in the loose case (Figure 2).

5.3 All Conversions

iso_tight/llgamma/all_conv/plots/kinematics_pt.pdf

(a) p_{T} (events)

iso_tight/llgamma/all_conv/plots/kinematics_pt_nor

(b) p_{T} (normalised)

iso_tight/llgamma/all_conv/plots/kinematics_eta.pdf

(c) $|\eta|$ (events)

iso_tight/llgamma/all_conv/plots/kinematics_eta_no

(d) $|\eta|$ (normalised)

Figure 6: All photons, tight isolation.

6 Discussion

6.1 Shape Agreement in p_{T}

Across all scenarios and isolation working points, the normalised p_{T} distributions show excellent shape agreement between data and MC. The ratio panels are consistent with unity within statistical uncertainties over the full p_{T} range. This confirms that the p_{T} spectrum of the selected photons is well modelled by the Sherpa simulation and that the event weight calculation (after the EVNT-level fix) is correct up to the overall normalisation factor discussed in Section 3.4.

6.2 Photon Conversion Rate Discrepancy

A striking observation is the significantly better data–MC agreement for **converted** photons (Figures 2, 5) compared to **unconverted** photons (Figures 1, 4), particularly in the $|\eta|$ distributions.

This is understood as a consequence of a known **TRT barrel tracker problem in Run 3**. The Transition Radiation Tracker (TRT), located in the barrel region ($|\eta| \lesssim 1.0$), plays a key role in classifying photons as converted or unconverted. In Run 3, a malfunction in the TRT barrel causes:

- A **higher fraction of photons classified as unconverted in data** than in MC, because the TRT is less efficient at reconstructing conversion tracks.
- Conversely, a **lower rate of converted photons in data** compared to MC.

A correction was previously applied that reclassifies TRT-converted photons as unconverted, but results from 2024 data show that this correction is **insufficient**. The residual mis-classification leads to:

- An *excess* of unconverted photons in data relative to MC, appearing as a normalisation deficit in the unconverted MC prediction.
- Better agreement for *converted* photons, because this category is less affected by the TRT barrel issue (endcap conversions are reconstructed differently).

This also explains why the $|\eta|$ shape discrepancy is concentrated in the barrel region ($|\eta| < 1.0$), where the TRT barrel is active.

6.3 Effect of Tight Isolation

Applying tight isolation instead of loose isolation slightly improves the overall normalisation agreement, as expected from the rejection of Z+jets background events in data. The shape discrepancy in $|\eta|$ for unconverted photons persists, however, confirming that it is not a background effect but rather the conversion-rate issue described above.

6.4 Scale Factors

The plots include an "MC (no SFs)" curve (blue dashed) in addition to the "MC (with SFs)" curve (red). The scale factors applied are: lepton reconstruction and identification SFs (SF_{ℓ_1}, SF_{ℓ_2}) and photon isolation SF ($\mathrm{SF}_{\gamma,\mathrm{iso}}$). The difference between the two MC curves is small, indicating that the scale factors have a modest impact on the kinematic distributions.

7 Summary and Conclusions

1. **Normalisation bug identified and fixed**: The DAOD-level sum-of-weights used by the NTupleMaker framework only accounts for ∼36% of the true EVNT-level total, inflating MC weights by ∼2.8×. Replacing with AMI EVNT-level values resolves the over-prediction.

2. **Residual ∼20% deficit understood**: The remaining MC under-prediction is attributed to missing Z+jets background in MC, incomplete MC sample coverage (97.59%), and residual efficiency effects. This does not affect the shower shape analysis, which uses normalised distributions.

3. p_{T} **shapes in good agreement**: The normalised p_{T} distributions show excellent data–MC agreement across all scenarios, confirming the MC modelling is adequate for the shower shape correction.

4. $|\eta|$ **discrepancy linked to TRT barrel**: The unconverted photon $|\eta|$ distribution shows a deficit in MC in the barrel region, while converted photons agree much better. This is attributed to a known TRT barrel tracker problem in Run 3 that alters the photon conversion classification rates between data and MC. A previous correction (reclassifying TRT-converted as unconverted) is insufficient for 2024 data.

5. **Tight isolation improves agreement marginally**: Tightening the isolation requirement reduces background contamination and slightly improves the normalisation, but does not resolve the conversion-rate discrepancy.

Overall, the kinematic distributions validate the use of the $Z \to \ell\ell\gamma$ sample for the cell-energy reweighting study. The p_{T} shape agreement ensures that the shower shape corrections are not biased by kinematic mis-modelling. The conversion-rate discrepancy motivates performing the shower shape analysis separately for unconverted and converted photons, and careful attention to the $|\eta|$-binned corrections.